

Motor Modeling

디지털제어 및 신호처리 - 2조

유민결 유채은 김솔

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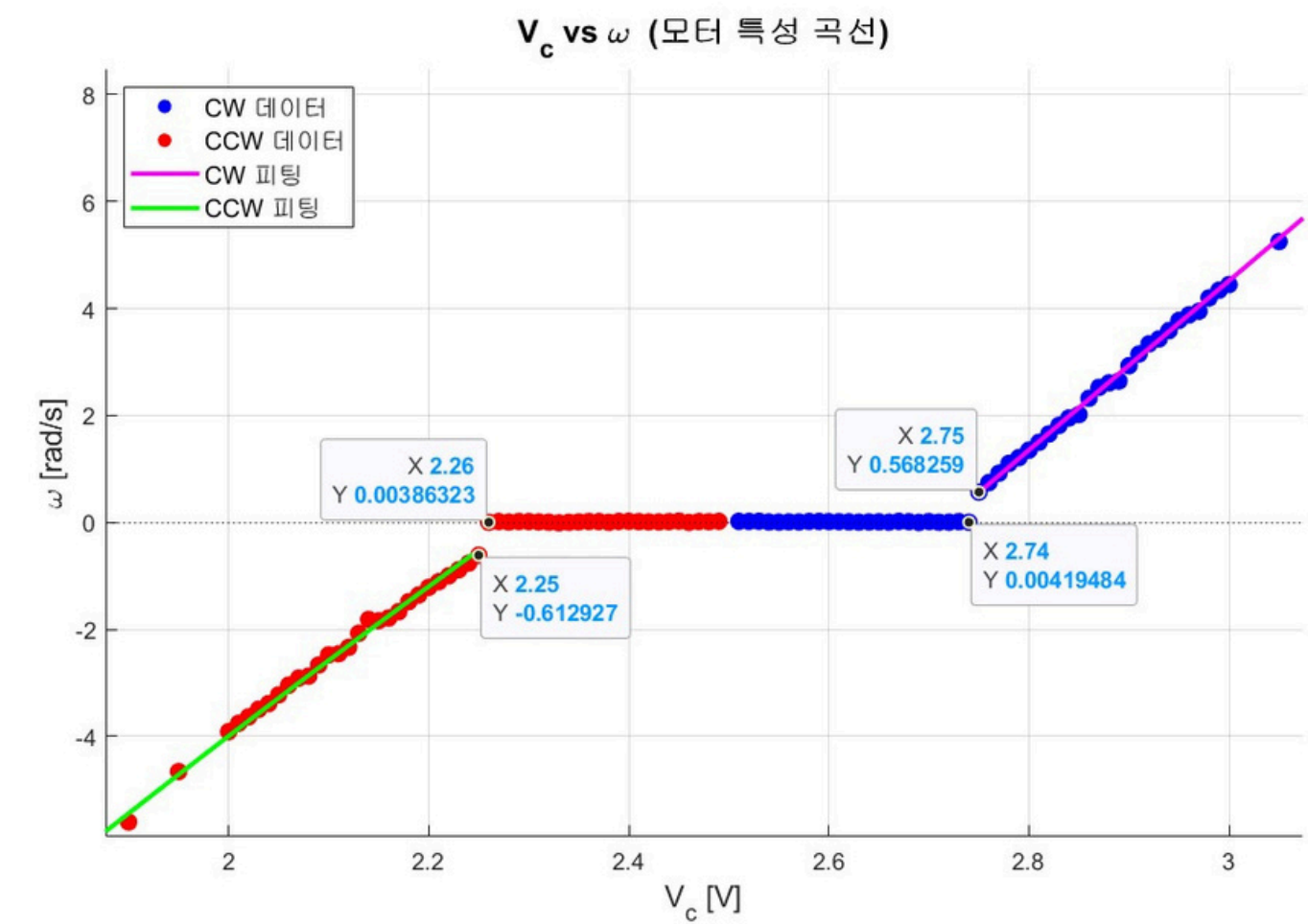
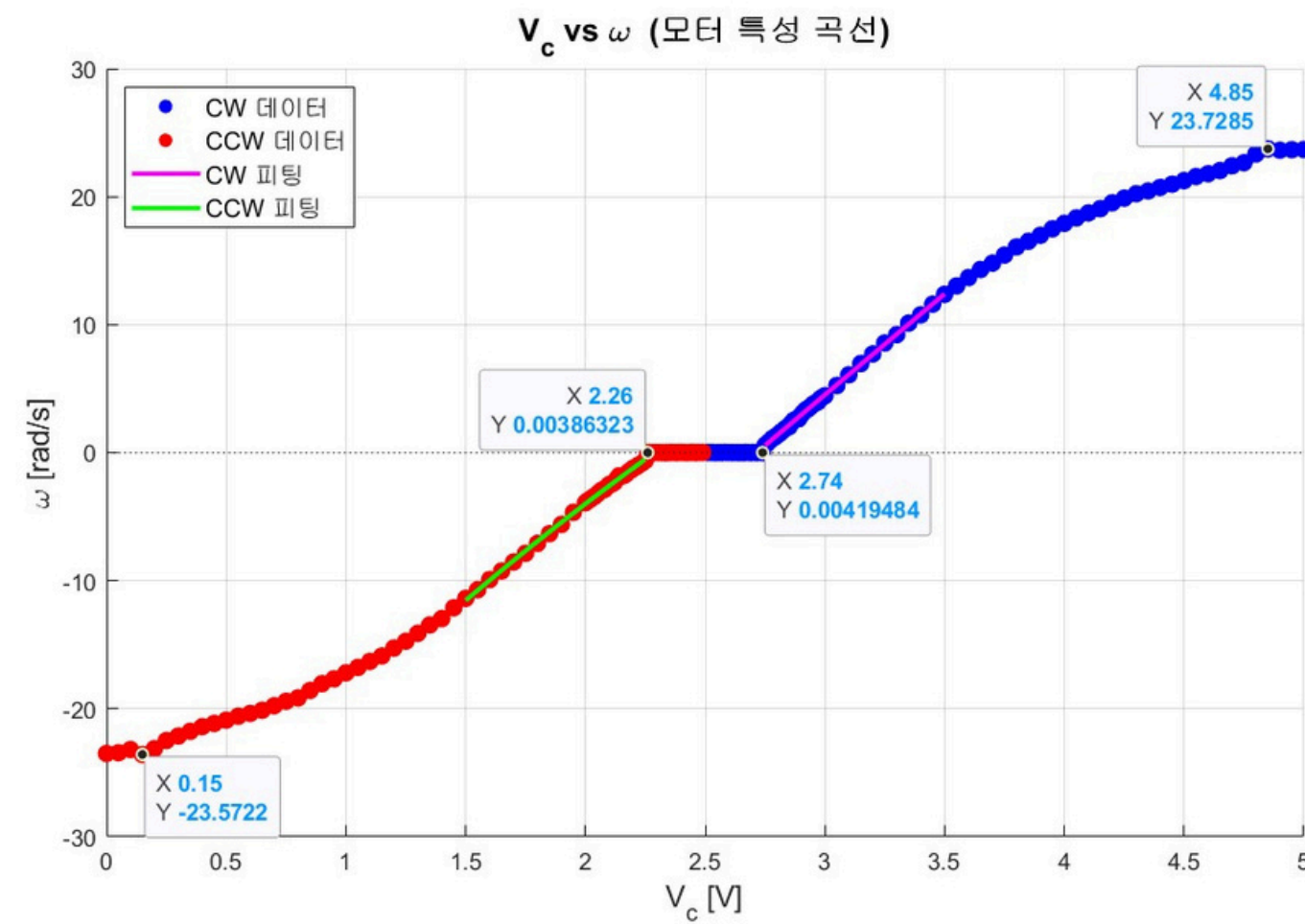
02. Dynamic Motor Modeling

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1. Static Motor Modeling

Deadzone 및 Saturation 확인

- $2 \leq V_c \leq 3$: 0.01V 간격
- $V_c < 2$ or $V_c > 3$: 0.05V 간격으로 data 수집



2차로 fitting

1. Static Motor Modeling

Linearization

- 선형화 범위 : 1.5[V] (CCW) $\leq V_{c_sat} \leq$ 3.5[V] (CW)로 설정
- 입력전압 범위 : 전체 ($-2.5[V] \leq V_{cmd} \leq 2.5[V]$) 사용한다고 설정

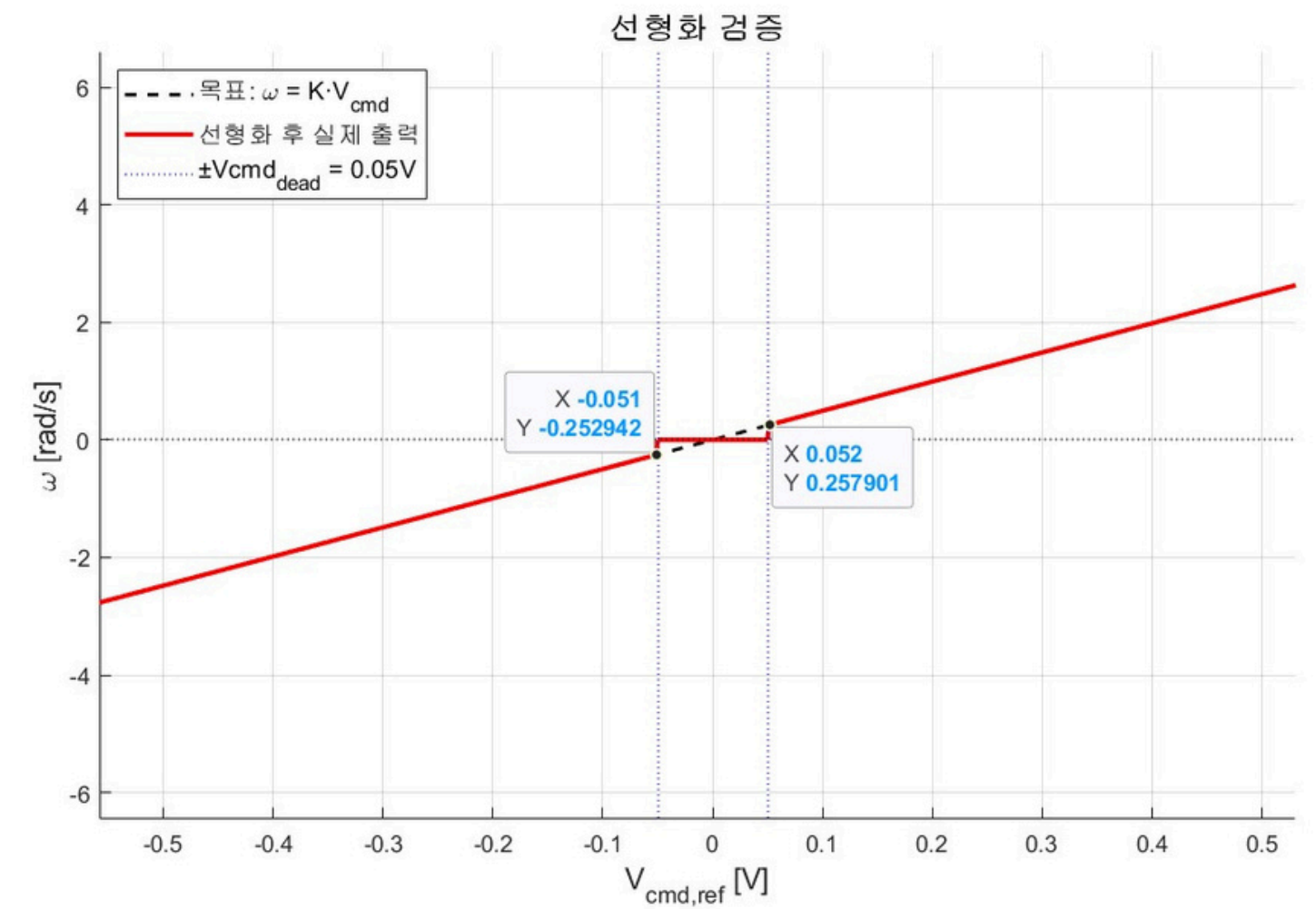
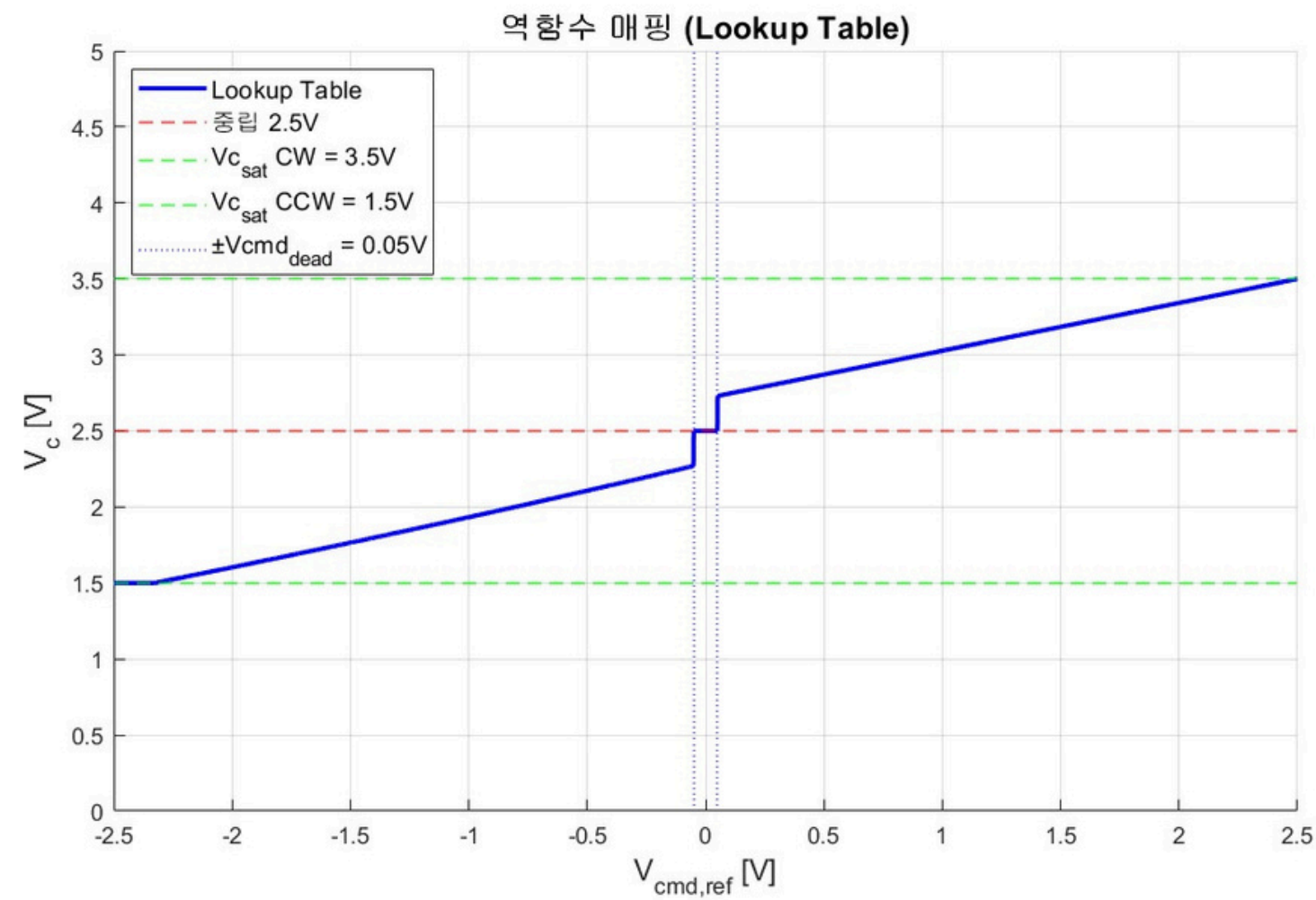
Vcmd [V]	-2.5	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5
Vc [V]	1.5	1.7	1.9	2.1	2.3	2.5	2.7	2.9	3.1	3.3	3.5

Mapping 관계

1. Static Motor Modeling

Linearization

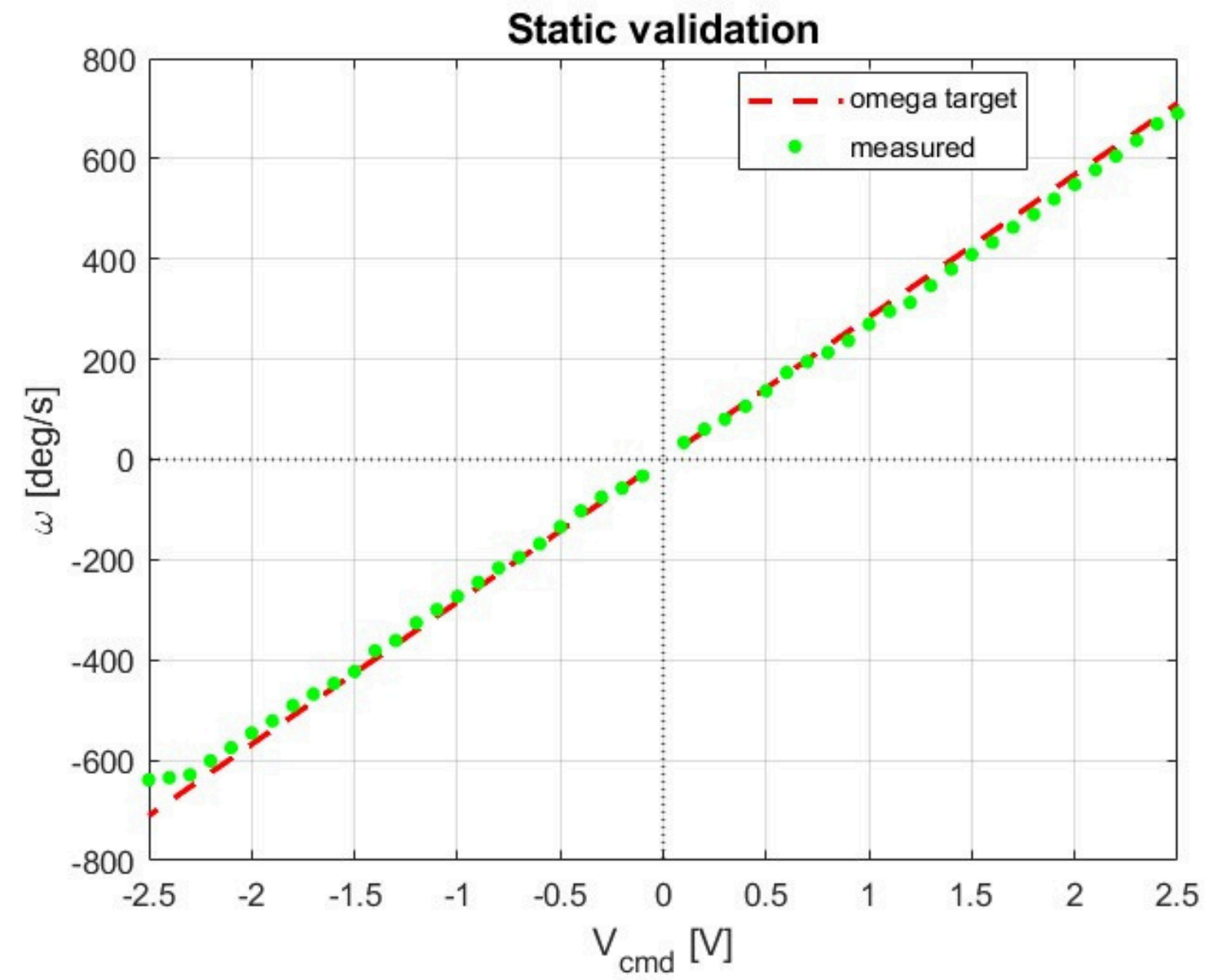
- 역함수 매핑 그래프



- $k = 4.9596$ [(rad/s)/V]
- V_{cmd} , Deadzone
(CW) : 0.052[V], (CWW) : -0.051[V]

1. Static Motor Modeling

Static Validation



1. Static Motor Modeling

Slewrates Limit 확인 및 amplitude A의 구간 추정

$$5 \cdot V_1 \leq A \leq \frac{\text{slewrates limit}}{k \cdot \omega_{BW}}$$

$$5 \cdot 0.05 \leq A \leq \frac{\text{slewrates limit}}{4.96 \cdot \omega_{BW}}$$

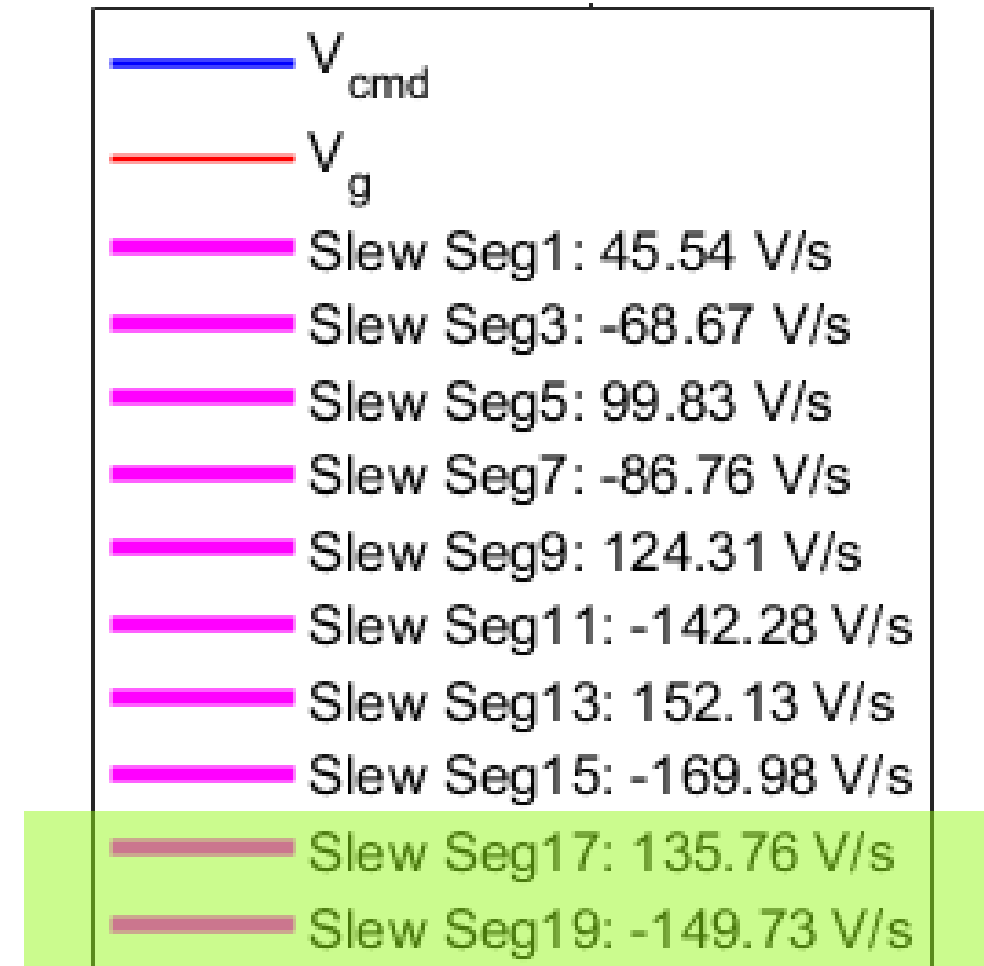
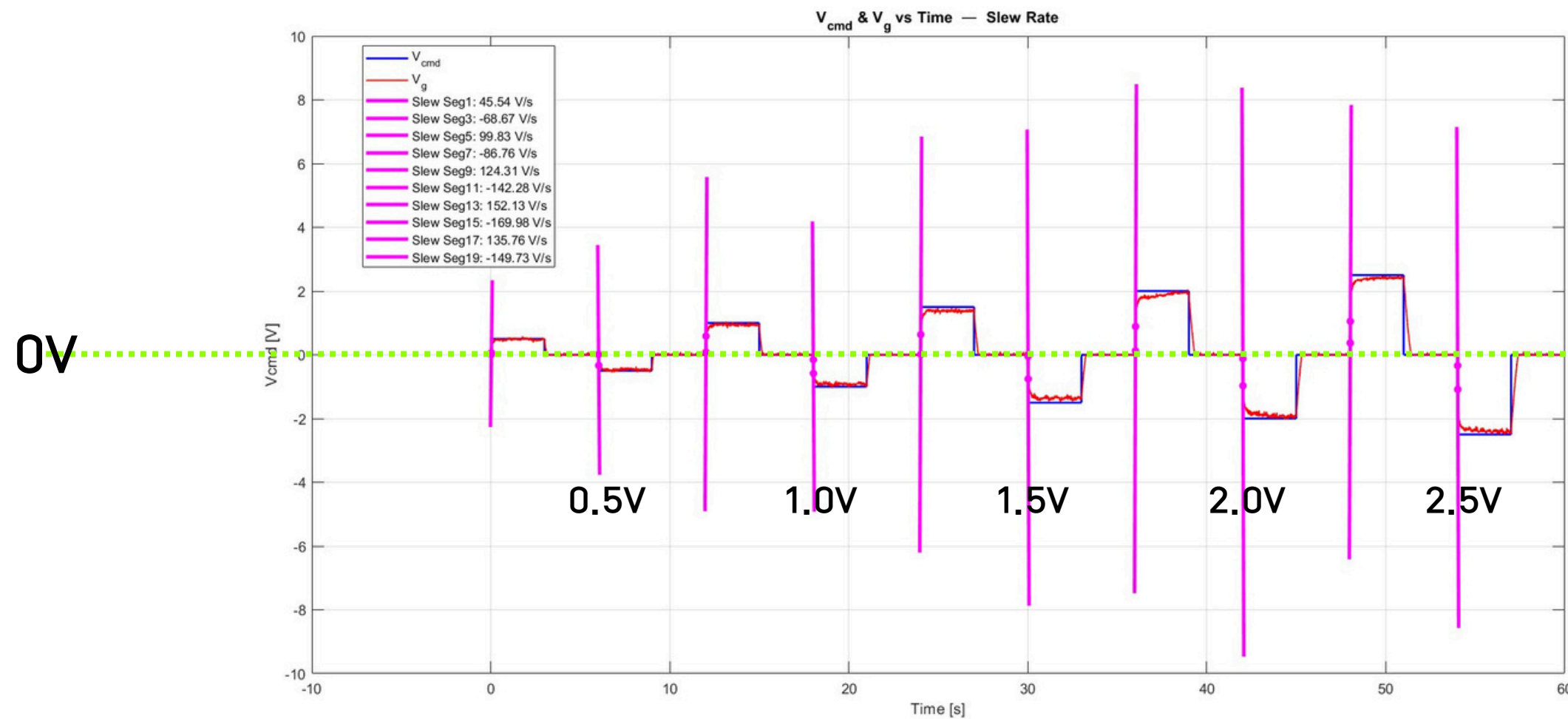
$$0.25 \leq A \leq \frac{\text{slewrates limit}}{4.96 \cdot \omega_{BW}}$$

A의 하한선 0.25 [V]

1. Static Motor Modeling

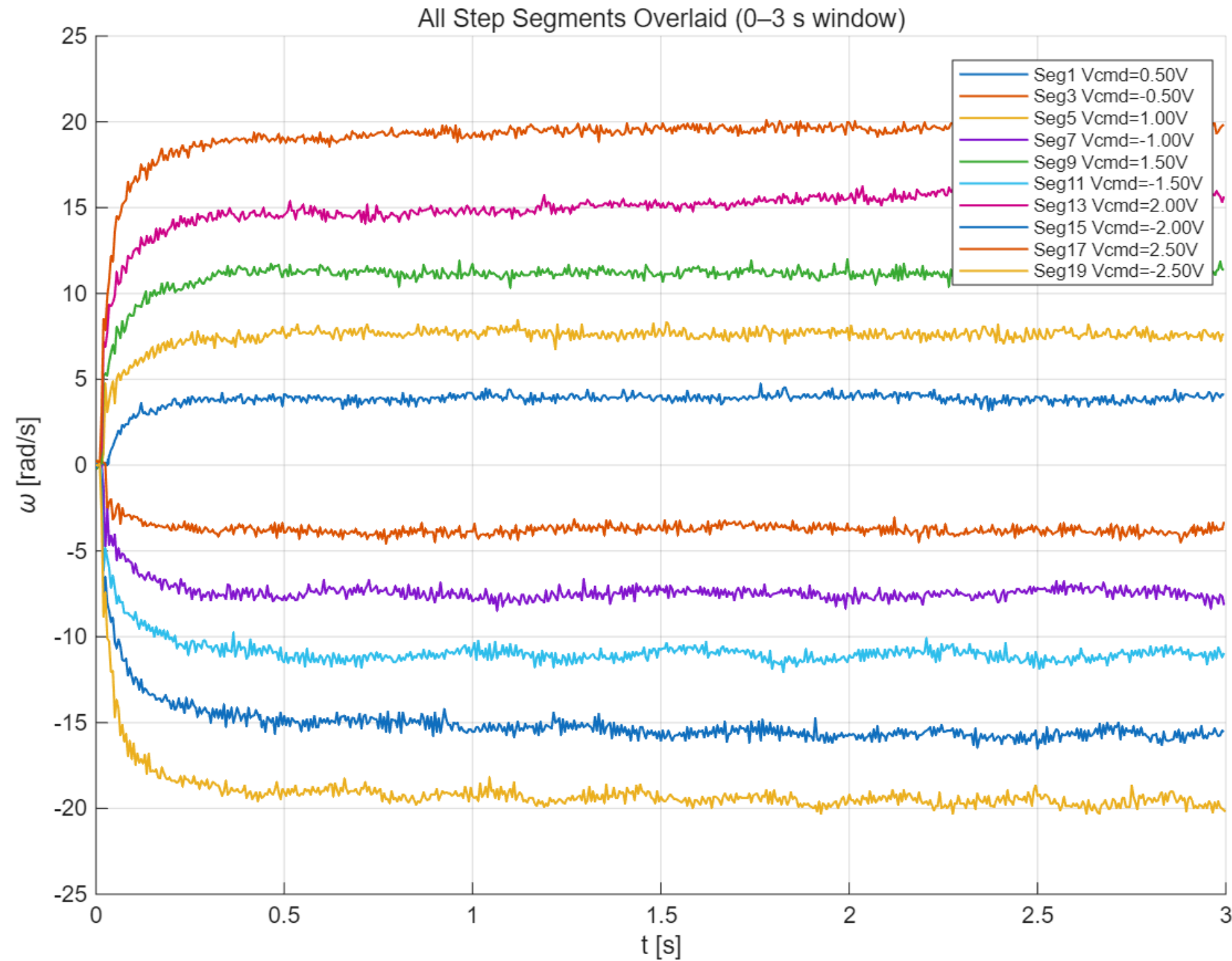
Slewrate Limit 확인

- 최대 입력전압 2.5V까지도 큰 왜곡 없이 추종
- 2.5V로 넘어가는 구간에서 기울기값이 약간 감소



1. Static Motor Modeling

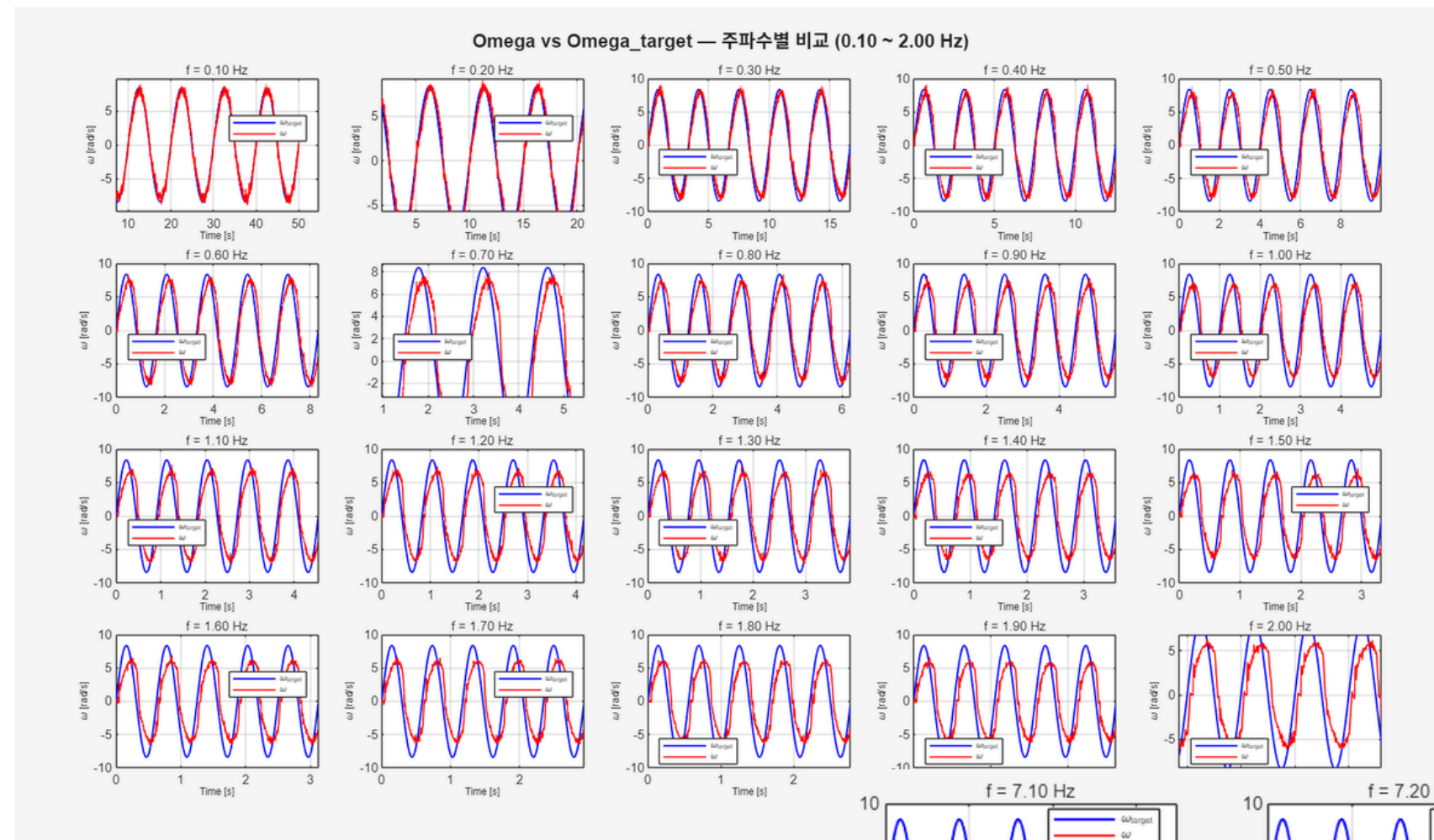
Slewrate Limit 확인



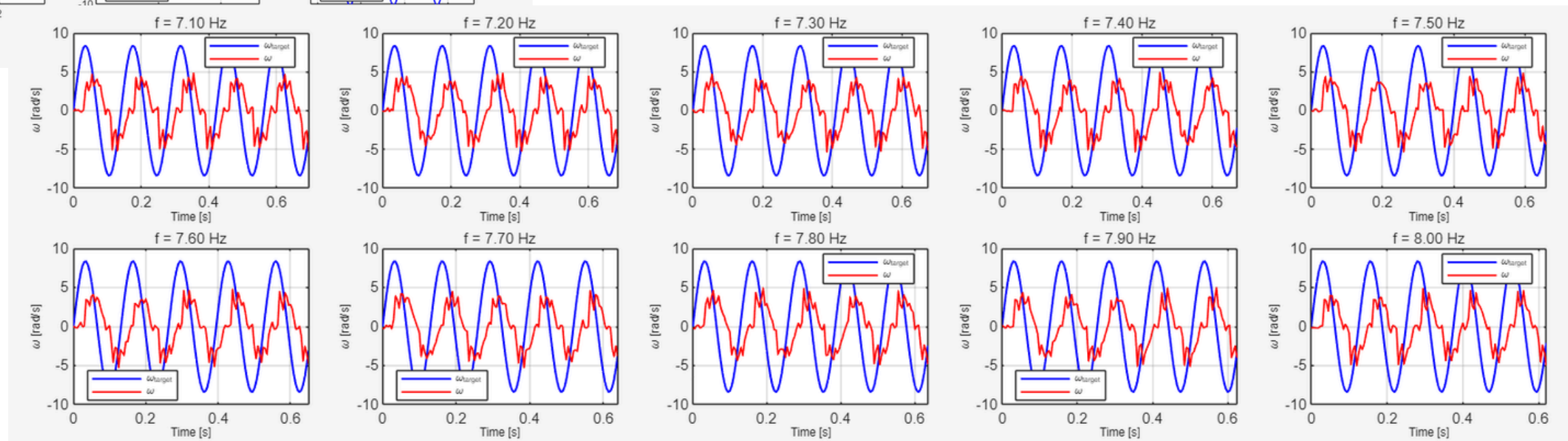
모든 구간에서의 그래프를 한번에 plot
기울기 간 명확한 차이 보기 어려움
→ slewrate limit이 있다고 보기 어려움

2. Dynamic Motor Modeling

Frequency sweep



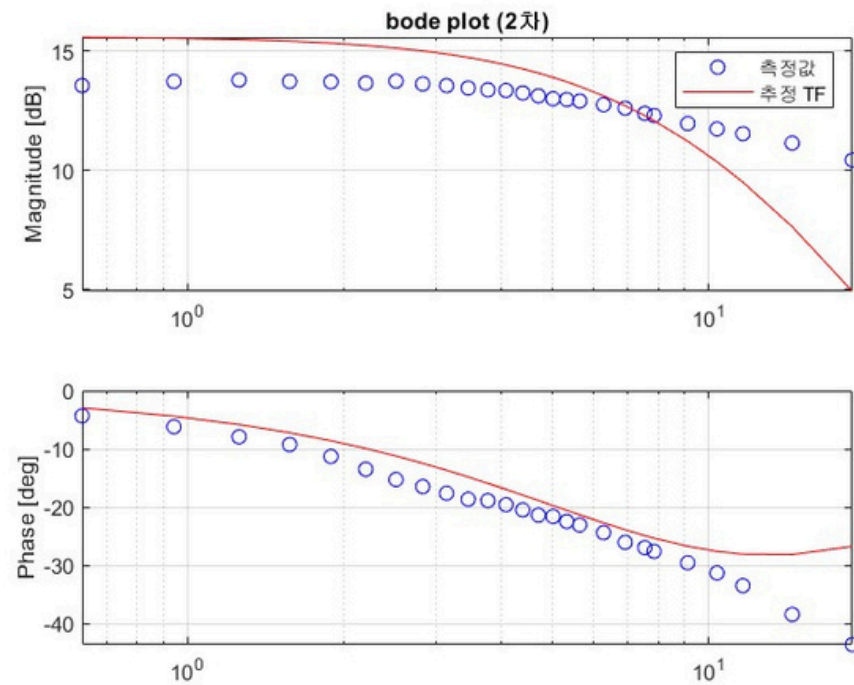
약 5.5 [V] 부근부터 파형이 깨진다고 봄



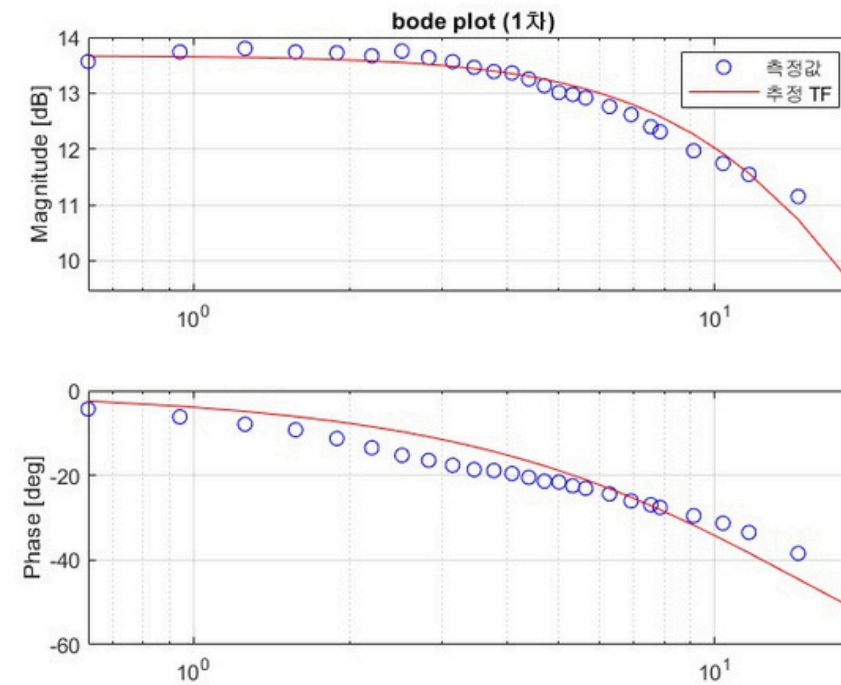
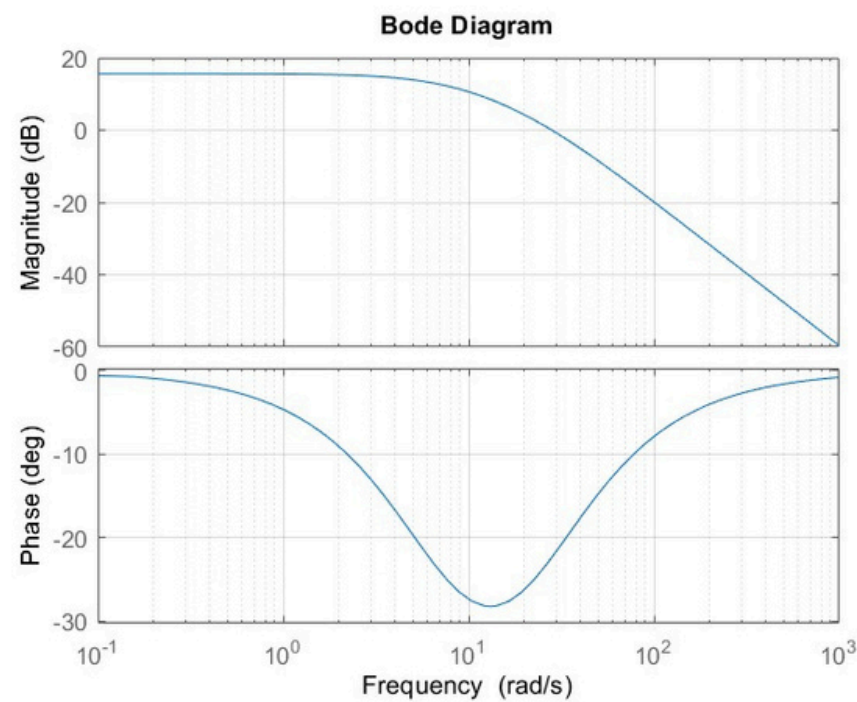
2. Dynamic Motor Modeling

Frequency sweep 및 모터 전달함수 추정

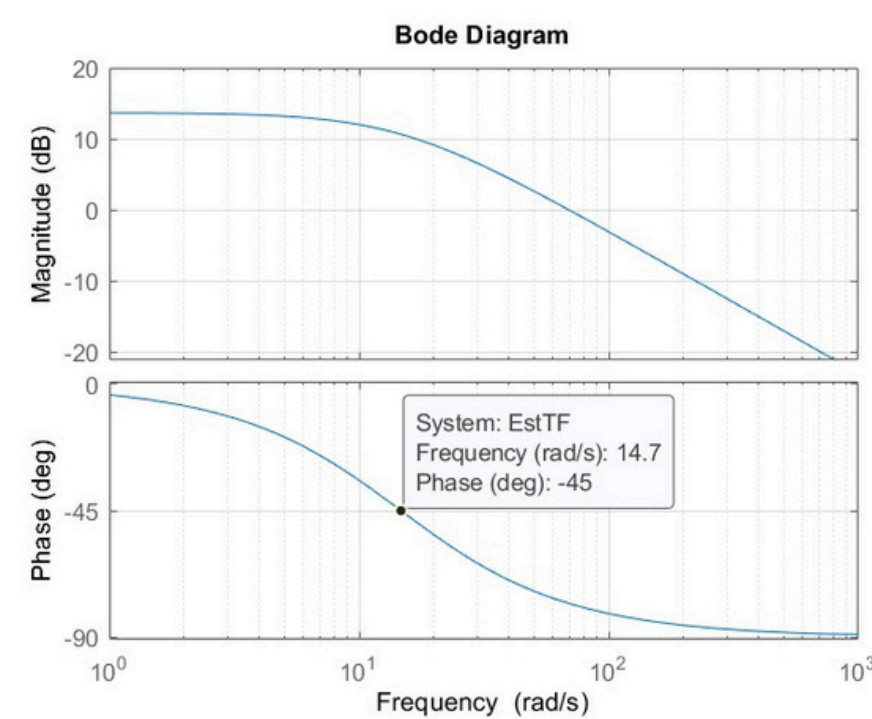
- Input freq: 0.1 ~ 3.0 [Hz]
- Amplitude: $V_{cmd} = 2.0[V] \rightarrow V_c = 3.3[V]$



2차



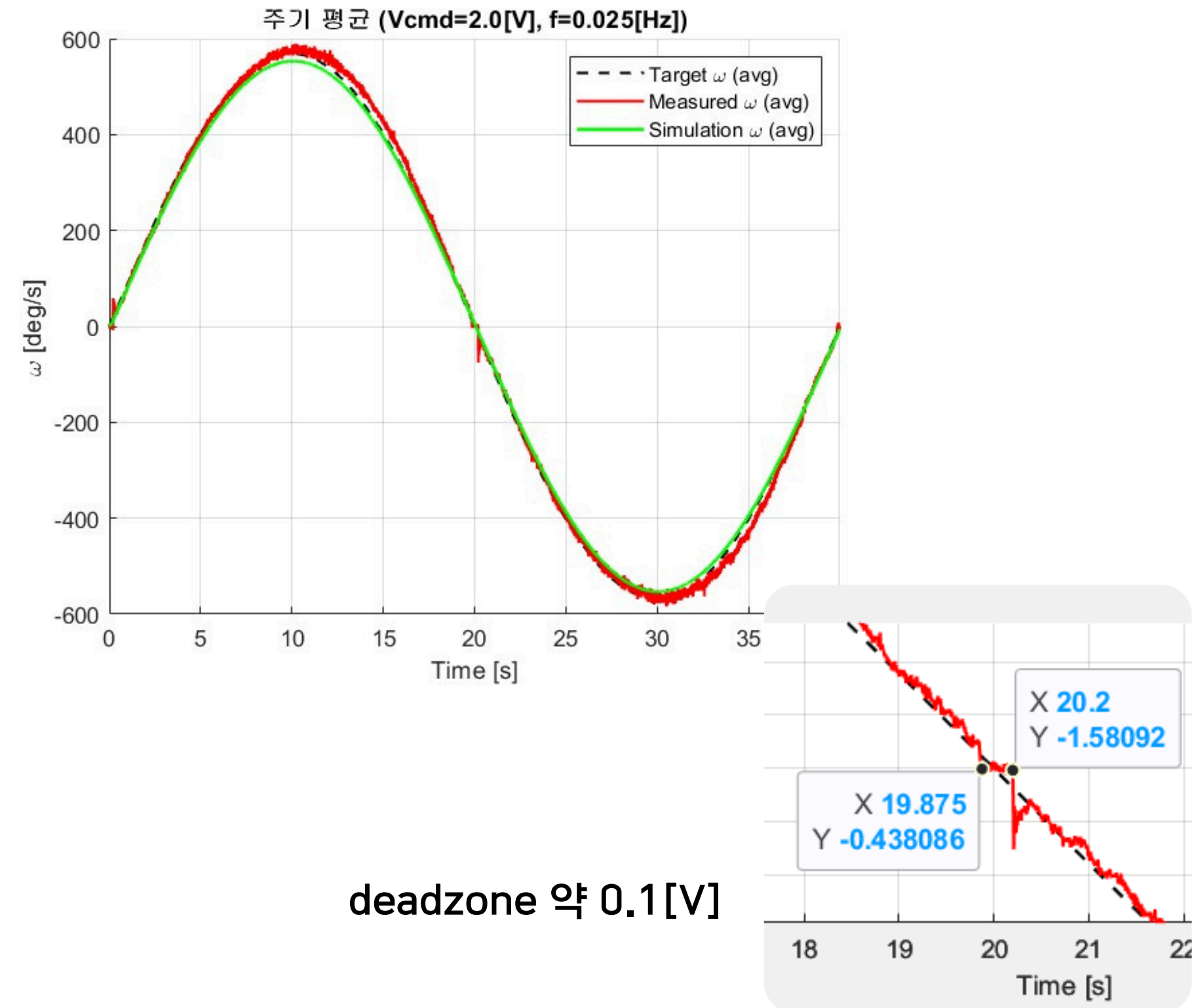
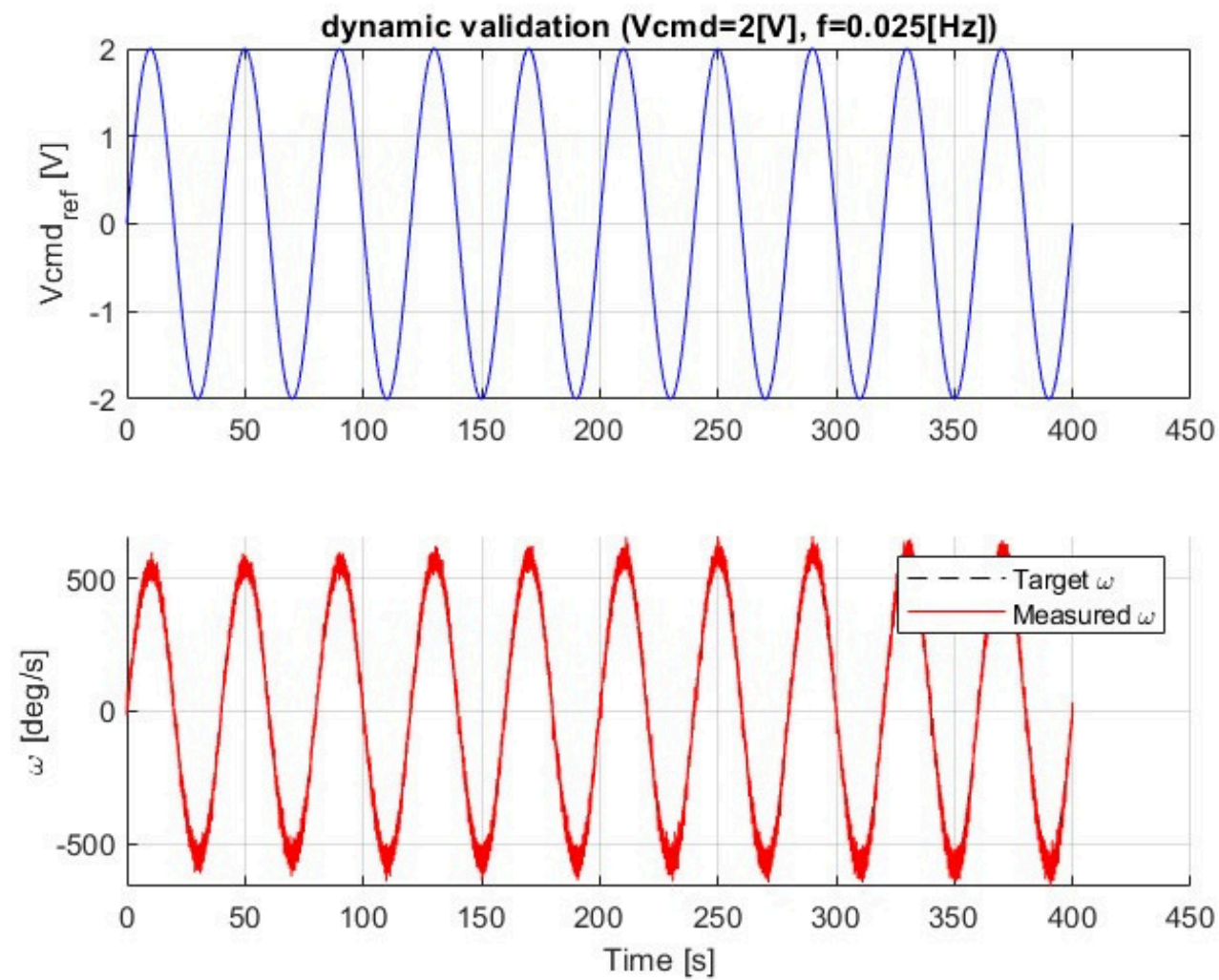
1차



$$G_m(s) = \frac{70.96}{s + 14.7}$$

3. Model Validation

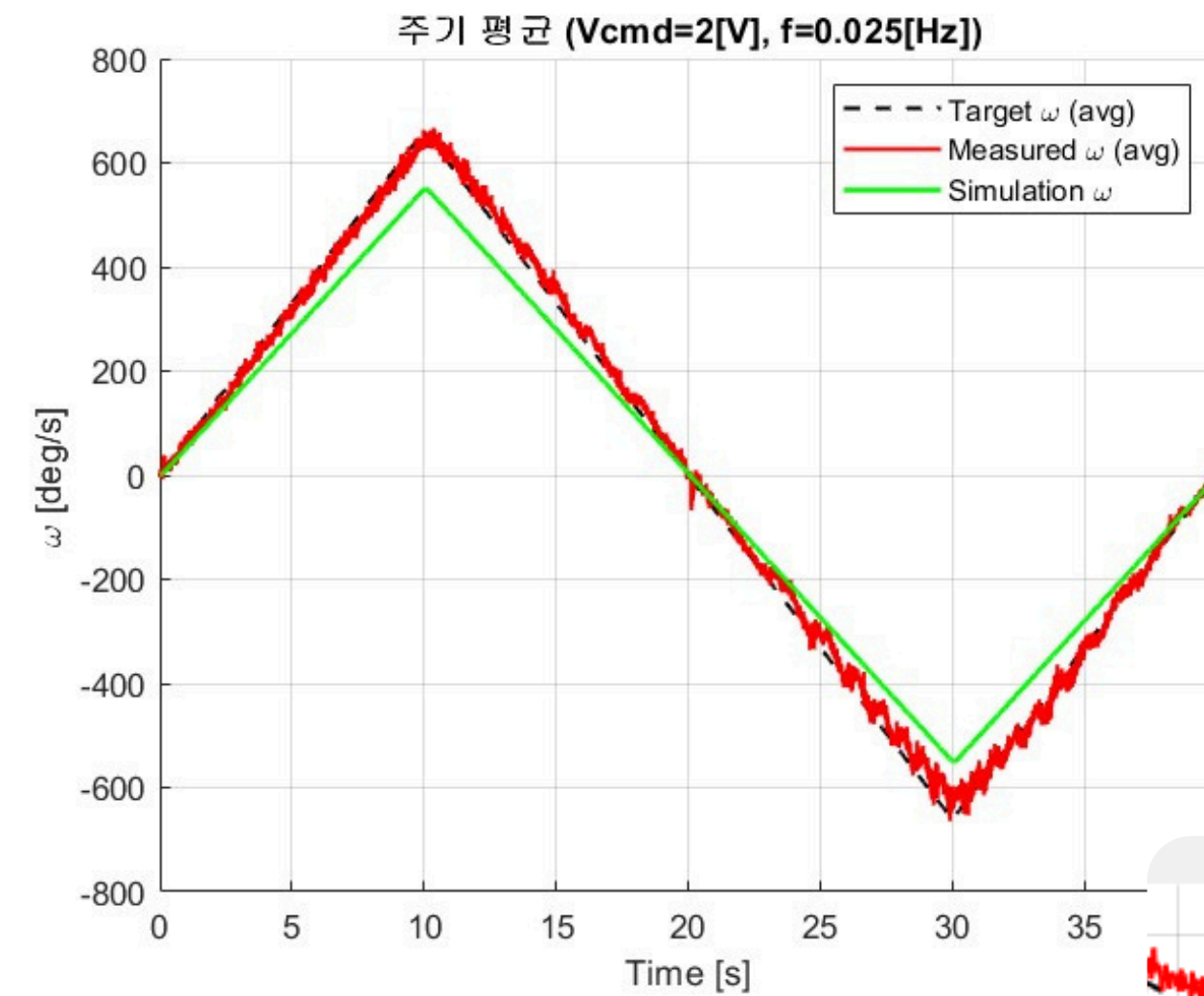
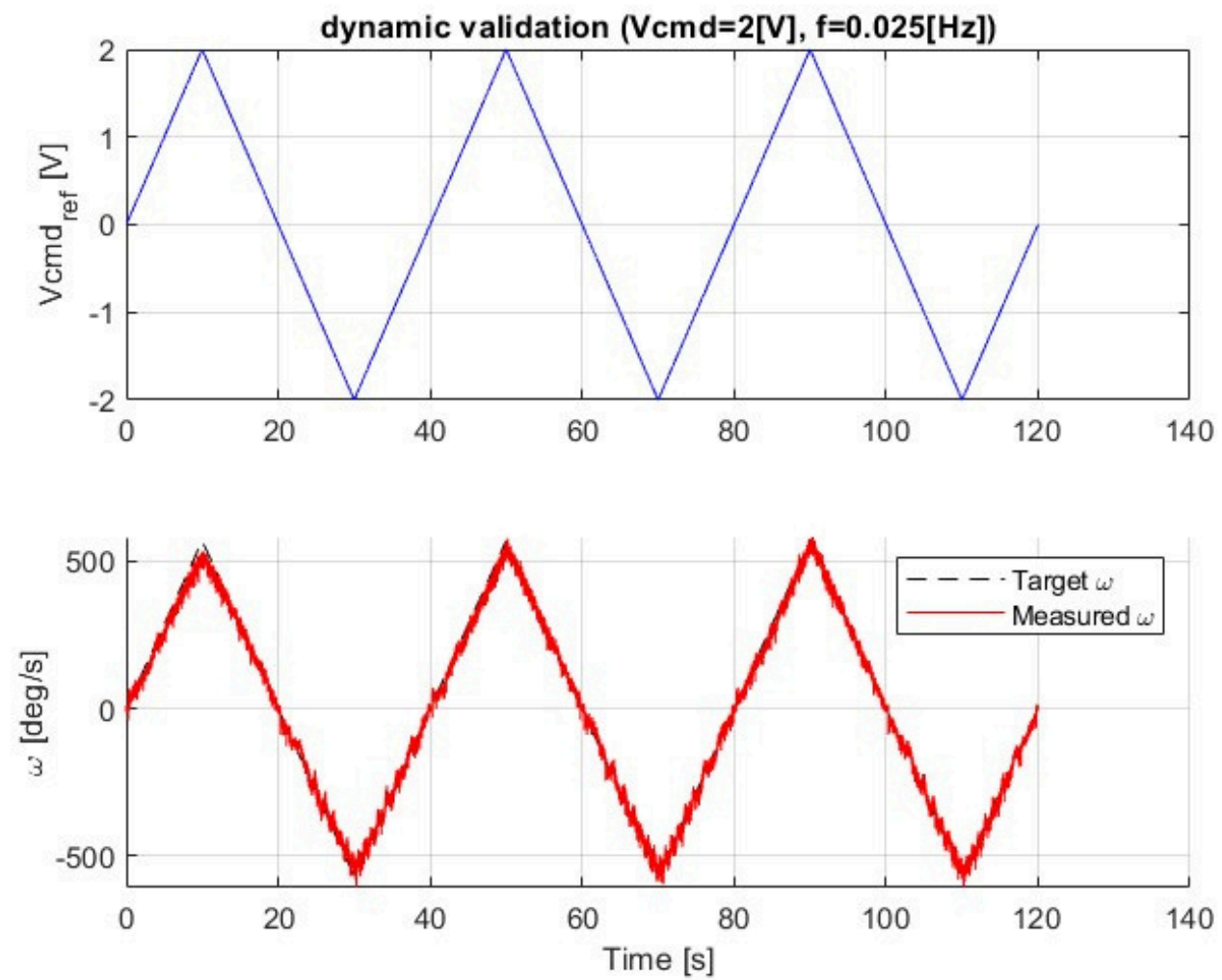
- 정현파 validation



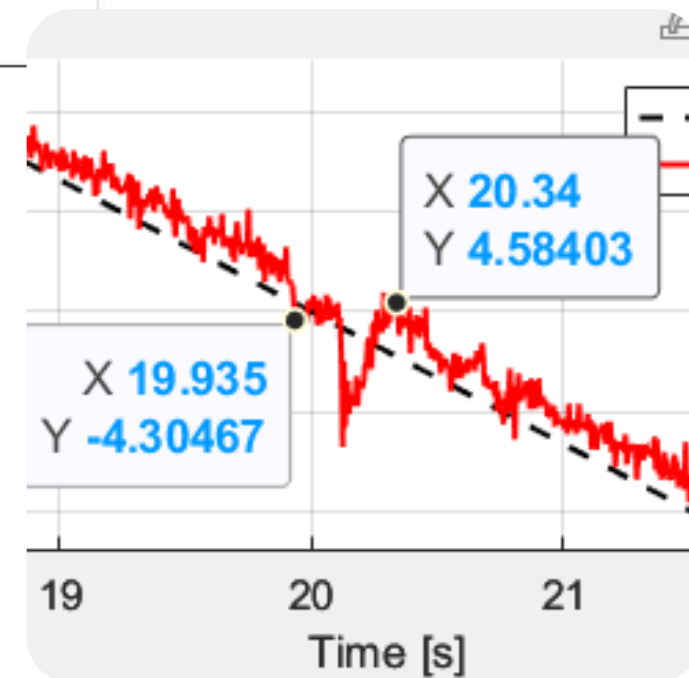
deadzone 약 0.1[V]

3. Model Validation

- 삼각파 validation



deadzone 약 0.08[V]



감사합니다